

CMSC 5033, Concepts of Artificial Intelligence

Dr. Shuai Zhang

Spring 2025

Instructor Contract information

Office Location MCS 127
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Office Hours MW 10:30 ~ 11:30, 15:30 ~16:30, or by appointment

General Course Information

Prerequisites CMSC 3613, Algorithms & Adv. Data Structures

Course Description This course introduces basic principles and techniques that serves as the most fundamental knowledge for understanding artificial intelligence (AI). We will learn the topics of problem solving by search, game playing, probabilistic reasoning, knowledge representation and some related machine learning or natural language processing applications.

Objectives Upon completion of this course, students will be able to

- demonstrate fundamental understanding of basic AI principles
- apply AI solutions that require modeling, inference, and learning for problem solving
- improve programming proficiency by developing applications in an 'AI language'
- envision the trend in AI study

Textbook Stuart Russell, Peter Norvig. *Artificial Intelligence: A Modern Approach*, 3rd Edition. Pearson, 2010.

Coursework Requirement

Attendance Attendance will be self-reported biweekly on D2L (learn.uco.edu). Learners need to follow the given report template.

Assignment Assignments should be completed **individually**. Intermediate procedures are required to demonstrate the correct operation of the assigned tasks.

Library Report A summary of a research paper related to the specified computer science topics is required to form basic ideas and practices of research approaches.

Project We will have a **term project** on an adversarial game. Each group can have two students. You need to join a group on D2L to be able to see the project.

Submission Submissions should be through D2L by the designated time.
Before submission, please make sure all the submission requirements are met. If D2L is not available, you can email the instructor your submission by due date, and then resubmit it on D2L when it's available (even if it's after the due time).

Due Date All assignments, report and project are due at the end of the day specified on D2L. Late submission won't be accepted unless it's with the instructor's permission (but no late more than one week after the specified due date).

Test Two tests will be given throughout the semester. The weight of each test is: 40% for the Midterm Exam, and 60% for Final Exam. The total weight of tests in the final grade is specified in the Grading Policy section below.

Grading Policy Weights

Tests	60%
Assignments	12.5%
Projects	20%
Report	2.5%
Attendance	5%

Grade Letter

Range of the final grade	Guaranteed term grade
90 and up	A
80 up to 89.99	B
70 up to 79.99	C
60 up to 69.99	D
below 60	F

Academic Dishonesty A student needs to demonstrate a high standard of individual honor in the scholastic work, and to protect your work from unauthorized access.
Any proven allegation of academic dishonesty will result in the offending student receiving **a grade of zero for the corresponding coursework**. Two blatant offenses will result in **a grade of F for the class and further disciplinary actions**.

Other Classroom Rules

Student Conduct & Discipline UCO respects students' rights as citizens and member of this community, and also needs to ensure students do not overlook their civic and social responsibilities. The Code of Student Conduct is fundamental to the normal activities at UCO for these purposes, and it can be accessed online at:
<https://www.uco.edu/offices/student-conduct/files/codeofconduct-webfile.pdf>

Disability Accommodation Students registered with the UCO Office of Disability Support Services may request accommodation of the prohibition and must present a copy of the DSS letter to the instructor. Students with disabilities who require accommodations may contact Disability Support Services with information at
<https://www.uco.edu/student-resources/dss/>

Classroom Order Cell phones and pages should be muted or turned off in class.
A student with any behaviors in violation to the Code of Student Conduct (listed above), could be asked to leave the classroom.

Privacy Due to limitations on the disclosure of personally identifiable information under certain **federal privacy laws**, students are not permitted to record class sessions or allow non-students to view online class sessions. Sharing links of class videos or add class videos to a public list is also prohibited.

Student Info Sheet You can find the student sheet at:
<https://www.uco.edu/academic-affairs/files/student-info-sheet.pdf>
Many useful information is provided for on-campus life.

Course Outline (Tentative)

Week	Topics, Holidays & Exams	Coursework Released
Jan 13 – Jan 17	Syllabus and Introduction	
Jan 20 – Jan 24	MLK day (No class on Jan 20) Uninformed Search	Graduate Lib Report
Jan 27 – Jan 31	Uninformed Search (cont'd); Informed Search	Assignment 1
Feb 03 – Feb 07	Informed Search (cont'd); Project Overview	Graduate Term Project
Feb 10-Feb 14	Informed Search (cont'd); Search Graph	Assignment 2
Feb 17 - Feb 21	Search Graph (cont'd); Adversarial Search	Assignment 3
Feb 24 – Feb 28	Adversarial Search (cont'd); Midterm Review	Assignment 4
Mar 03 – Mar 07	Local Search; Midterm Exam (Wednesday, Mar 06)	
Mar 10 – Mar 14	Knowledge Representation	
Mar 17 – Mar 21	Spring Break (No class on Mar 17 and Mar 19)	
Mar 24 – Mar 28	Knowledge Representation (cont'd); First Order Logic	
Mar 31 – Apr 04	First Order Logic (cont'd)	
Apr 07 – Apr 11	Probability Reasoning; Bayesian Network	Assignment 5
Apr 14 – Apr 18	Bayesian Network (cont'd)	Assignment 6
Apr 21 – Apr 25	Advanced Topics	
Apr 28 – May 02	Class Review	
May 05 – May 09	Graduate Final Exam (13:00 ~ 14:50, Wednesday, May 07, Comprehensive)	

Notice: these descriptions and timelines are subject to change at the discretion of the instructor, based on course progress.